Wonguk Lee

Ph.D. Student in Management Information Systems
Business and Technology Management, KAIST
Advisor: Professor Chul Ho Lee | lwg5253@kaist.ac.kr

Education

Korea Advanced Institute of Science and Technology

Ph.D. Student, Business and Technology Management; major in Management Information Systems.

Korea Advanced Institute of Science and Technology

M.S., Business and Technology Management; major in Management Information Systems.

Handong Global University

B.B.A. in Business Administration; B.A. in Law (double major).

Appointments and teaching

Chief Data Officer, TalentSeeker

February 2024–present. Full-time C-level role. Lead data ontologies, multi-agent systems, data-pipeline architecture, data engineering, and LLMOps for AI-enabled recruitment.

Lead Course Instructor in Data Science, Baekseok University

Second semester, 2026. English-medium course design and teaching in Python, pandas, data preparation, visualization, statistics, prediction, and responsible generative AI use.

Upcoming presentations

CES 2027

Scheduled to attend as a C-level speaker in my role as Chief Data Officer at TalentSeeker.

Workshop on Information Technology and Systems

Protecting the Investor, Punishing the Retailer: accepted for presentation at WITS; presentation forthcoming.

Research interests

Mobility and access to services; human–AI interaction and mental healthcare; digital platforms and AI governance; technological innovation and knowledge flows.

Honors and awards

Excellent Paper Award, Korea Society of IT Services, for the Tashu and public-transit study. Outstanding Research Idea Award, BTM Seminar, KAIST, for the study of AI advisor withdrawal in P2P lending. Excellent Academic Book selection, National Academy of Sciences, Republic of Korea (2025).

Published and forthcoming

Sectoral Heterogeneity in the Digital Transformation-Financial Performance Nexus: Evidence from Korean Manufacturing Firms

Junehwan Kim†, **Wonguk Lee**†, and Euiseok Kim
† Equal contribution.
Accepted · Forthcoming in Asian Journal of Technology Innovation

Thought-Bubbles in Therapy: The Effect of Conversational AI's Transparency in Mental Health Contexts

Wonguk Lee, Donggyu Min, Antino Kim, and Chul Ho Lee
ICIS 2026 Proceedings, Healthcare and Wellbeing, Paper 19. Completed research paper.

Proceedings

Transformation of Business Models in Metaverse Era: Through a Systematic Literature Review and Bibliometric Analysis

Temitayo Shenkoya, **Wonguk Lee**, and Euiseok Kim (2024).
Journal of Information Technology Services, 23(6), 1–17.

DOI

A Study on the Flow of Technological Knowledge in Korean Industries: Focusing on Inter-industry Patent Citation Analysis Based on Pavitt's Taxonomy

Wonguk Lee and Euiseok Kim (2024).
Journal of Korea Technology Innovation Society, 27(3), 431–454.

DOI

A Study on the Intention to use Personal Mobility Services: Focused on the SOR (Stimulus-Organism-Response) Model

Wonguk Lee and Heetae Yang (2022).
Information Systems Review, 24(2), 67–88.

DOI

Books and chapters

Generative AI as a Creative Collaborator

Lee, W. (2026). In AI Transformation Era: Innovation and Challenges of Korean Industries Highlighted by KAIST. Yulgok Publishing.

AI Era: Structure and Sustainability of the Korean Workforce Ecosystem

Co-author. Selected as an Excellent Academic Book by the National Academy of Sciences, Republic of Korea (2025).

Working papers

Formation and Dissipation of the Hypernetwork Effect: Transitional Complementarity between Learning and Participation in AI-Driven Networks

Wonguk Lee and Euiseok Kim
Under revision · Second round

Primary Token Issuance and Liquidity-Segmented Returns in Crypto Markets

Kyeong Hun Kim, **Wonguk Lee**, and Byung Hwa Lim
Under revision · Minor revision

Understanding corporate digital entrepreneurship in financial institutions: The roles of AI adoption and entrepreneurial orientation

Heetae Yang and **Wonguk Lee**
Submitted

Understanding Personal Mobility Device Adoption Through the S–O–R Framework: A Multi-Group Analysis of Users and Non-users

Wonguk Lee and Heetae Yang
Under review

The Effects of Public Bike-Sharing (Tashu) and Public Transit Integration on Local Festivals and Consumer Spending

Kyusung Kim, **Wonguk Lee**, Haeun Jang, and Hyungu Jang
Under revision · Excellent Paper Award, Korea Society of IT Services

The Role of NFTs in the Digital World: Catalyzing Innovation in the Creative Industry through a Patent-Inspired Approach

Wonguk Lee and Euiseok Kim
Under review

Work in progress

Breadth to Enter, Relatedness to Stay: Technology Portfolios and the Limits of Firms' Green Diversification

Wonguk Lee and Euiseok Kim
Work in progress

Thought-Bubbles in Therapy: The Effect of Conversational AI's Transparency in Mental Health Contexts

Wonguk Lee, Donggyu Min, Antino Kim, and Chul Ho Lee
Work in progress; related ICIS 2026 proceedings version listed above.

Protecting the Investor, Punishing the Retailer: The Perverse Effects of Banning AI Autonomous Investment

Wonguk Lee, Gunsu Kim, and Chulho Lee
Work in progress · Accepted for presentation at WITS; presentation forthcoming.

Funded but Not Followed? Donor Relationships After Sponsor-Led Funding Completion

Wonguk Lee and Chul Ho Lee
Work in progress

AI Summary Design in Consumer Complaint Platforms

Wonguk Lee and Chul Ho Lee
Work in progress · Working title

Patents and applications

1. Ontology-Based Data Standardization for Human Resource Management

Korean patent 10-2891011 · Registered 20 November 2025. Application 10-2025-0100830 · Filed 24 July 2025. A related PCT international application has also been filed.

2. Ontology-Based Human Resource Analysis and Job Inference

Korean application 10-2025-0174900 · Filed 18 November 2025. Decision to grant received.

3. LLM-Based Ontology Data Processing for Human Resource Information

Korean application 10-2025-0176365 · Filed 19 November 2025. Decision to grant received.

4. Talent Search and Recommendation Based on Conversational Hiring-Needs Analysis

Korean application 10-2026-0110525 · Filed 17 June 2026. Under prioritized examination.

5. Resume Data Augmentation and Personalized Inference Using Relationship-Aware LLM Context

Korean application 10-2026-0082060 · Filed 7 May 2026. Under prioritized examination.

6. Road Determination Device and Method of Determining a Road

Korean application 10-2025-0209757 · Filed 24 December 2025

7. Recommended Speed Determination Apparatus and Method

Korean application 10-2025-0207816 · Filed 23 December 2025

8. Ontology-Based Natural-Language Query Processing and Interactive Candidate Filtering

Korean application 10-2025-0170399 · Filed 12 November 2025. Under prioritized examination.

9. Method and Apparatus for Evaluating Image Tags for AI Training

Korean application 10-2024-0067992 · Filed 24 May 2024

10. Method for Predicting Fracture of Pipe Due to Corrosion of Alloy and System Thereof

Korean patent application specification · 17 April 2024

Research and development projects

Smart Mobility Safety Services and Infrastructure for Autonomous Driving

MOTIE / KIAT, International Industrial Technology Cooperation | Project P0025238
2025 | Technical development and patent contribution
Developed OSM-supported road-detection and road-geometry algorithms and physics/XGBoost-based speed recommendations with KAIST and industry collaborators.

Public Bike-Sharing and Transit Integration for Local Mobility and Consumer Activity

Yuseong District Office and Daejeon Transportation Corporation support | Project lead
Secured and led the project; integrated Tashu bike-sharing, transit, Shinhan Card spending, and KT population-movement data to study first/last-mile connections, local festivals, and economic activity.

Personal Mobility Device Adoption among Users and Non-users

Behavioral research | Manuscript under review
PLS-SEM and multi-group analysis of 444 respondents, including 222 users and 222 non-users; examined perceived physical risk, trust, and adoption intentions.

AI Counseling Chatbot for Schizophrenia Care

Jeonbuk National University Hospital | Ongoing
Develop and study a free patient-facing LLM/RAG chatbot, focusing on perceived judgment, stigma-related engagement barriers, disclosure, and supportive interaction design.

Onto-LLM-Based Global Outbound Recruitment Service Development

TIPS R&D Program, Ministry of SMEs and Startups (MSS) | Aug 2025 – Jul 2027 | Ongoing
Contribute to ontology-based recruitment AI, Graph-RAG / Graph-of-Thought reasoning, and agent prototype research.

Open-Web-Based Automated Resume Collection and AI Recruitment Support System

Startup Growth Technology Development Program, TIPA / MSS | Aug 2024 – Jul 2025 | Completed
AI recruitment and data-pipeline development.

Multiple-RAG and Multidimensional Ontology-Based Recruitment sLLM

Didimdol R&D planning | 2026
Research and engineering design for a recruitment-specific small language model, evidence retrieval, ontology mapping, evaluation, and an end-to-end processing pipeline.

SF-Olive Smart Farm Platform for Olive Production

International Collaborative Technology Development | Planned project period Aug 2026 – Jul 2029
KAIST participating researcher role: data experimentation and bias-analysis support for a platform combining advanced sensing, blockchain, and edge AI semiconductors.

Research methods

Methods: GIS and OSM networks; physics-based modeling and XGBoost; PLS-SEM and multi-group analysis; experiments; panel-data analysis; agent-based simulation; patent analysis. Tools: Python, pandas, data engineering, ontologies, RAG, multi-agent systems, and LLMOps.